

APPLIED STATISTICS, PROBABILITY THEORY AND MATHEMATICAL STATISTICS

Attila Barta, Sándor Pecsora

General notion of random variables

Exercise 9.3

Check whether the following functions are cumulative distribution functions or not.

$$a) \quad F(x) := \begin{cases} 0, & \text{if } x < 1/2 \\ \frac{x-1}{x+1}, & \text{if } x \geq 1/2 \end{cases}$$

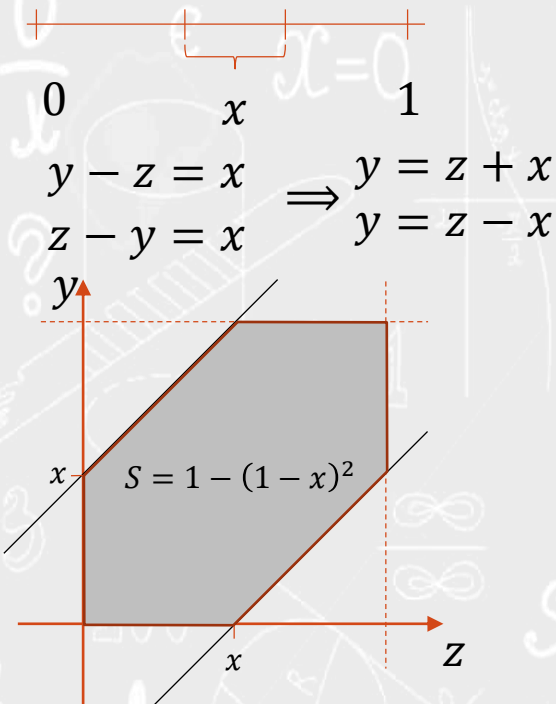
$$b) \quad F(x) := \begin{cases} 0, & \text{if } x < 1 \\ \frac{x-1}{x+1}, & \text{if } x \geq 1 \end{cases}$$

a) The $\frac{x-1}{x+1}$ is negative when $x = \frac{1}{2}$; $F(x)$ is not CDF.

b) $F(x)$ is a CDF

Exercise 9.5 (M1) Only Matlab

Find the cdf of the distance of two randomly chosen points of the $[0; 1]$ interval. What is the probability that the distance of the points is in the interval $\left[\frac{1}{2}; \frac{3}{4}\right]$?



$$1 - (1 - x)^2 = x(2 - x)$$

$$F(x) = P(\xi < x) = \begin{cases} 0, & \text{if } x \leq 0 \\ x(2 - x), & \text{if } 0 < x \leq 1 \\ 1, & \text{if } 1 < x \end{cases}$$

$$P\left(\xi \in \left[\frac{1}{2}; \frac{3}{4}\right]\right) = F\left(\frac{3}{4} + 0\right) - F\left(\frac{2}{4}\right) = \frac{3}{16}$$

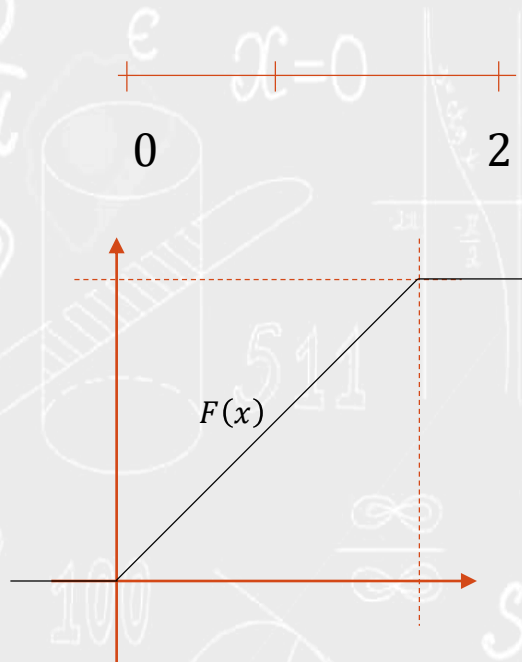
Exercise 9.5 (M1)

MATLAB solution

```
n = 10^4;  
c = 0;  
axis([0 1 0 1])  
hold on  
for i=1:n  
    x = rand();  
    y = rand();  
    if abs(x-y)>1/2 & abs(x-y)<3/4  
        plot(x,y, 'g')  
        c = c+1;  
    else  
        plot(x,y, 'r')  
    end  
end  
c/n  
3/16
```

Exercise 10.1 (M2)

A stick of length two meters is randomly broken into two parts. Find the cumulative distribution function (cdf) and the probability density function (pdf) of the length of the shorter part.



$$F(x) = P(\xi < x) = \begin{cases} 0, & \text{if } x \leq 0 \\ x, & \text{if } 0 < x \leq 1 \\ 1, & \text{if } 1 < x \end{cases}$$

$$\xi \sim u(0; 1)$$

$$f(x) = \begin{cases} 1, & \text{if } 0 < x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Exercise 10.1 (M2)

MATLAB solution

```
n = 10^5;
```

```
stick_1 = rand(1,n)*2;
```

```
stick_2 = 2-stick_1;
```

```
shorter = min([stick_1; stick_2]);
```

```
ecdf(shorter)
```

```
hold on
```

```
plot([0 1],[0 1],'r')
```

Exercise (Uniform distribution)

Let denote X the waiting time at a bust stop. The waiting time at a bus stop is uniformly distributed between 1 and 12 minute. That is $X \sim U(1,12)$.

a) The pdf of X is:

$$f(x) = \frac{1}{12 - 1} = \frac{1}{11}; \quad 1 \leq x \leq 12$$

b) The probability that the rider waits 8 minutes or less is

$$P(X \leq 8) = \int_1^8 f(x) dx = \frac{1}{11} \int_1^8 dx = \frac{7}{11}$$

c) The expected waiting time is $\mathbb{E}(X) = \frac{1 + 12}{2} = 6.5$

d) The variance waiting time is $\mathbb{D}^2(X) = \frac{(12 - 1)^2}{12} = \frac{121}{12}$

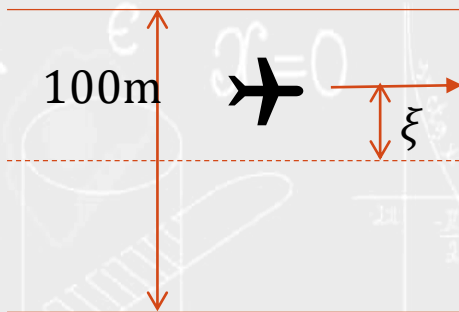
Exercise (Uniform distribution Matlab solution)

```
n = 10^5;  
x = rand(1,n)*11+1;  
p = length(x(x<8))/n  
x_mean = mean(x)  
x_var = var(x)
```


Exercise 13.5 (Normal distribution)

The air control informs the pilot of an aircraft about the altitude of the centre of the air corridor of height 100 metres where the aircraft should fly. The deviation, in metres, of the altitude of the aircraft from the given altitude is normally distributed with mean 20 and standard deviation 50. Find the probability that the aircraft flies under the air corridor, in the air corridor, above the air corridor.

$$\xi \sim \mathcal{N}(20, 50^2)$$



Above the air corridor

$$\begin{aligned} P(\xi > 50) &= 1 - P(\xi < 50) = 1 - P\left(\frac{\xi - 20}{50} \leq \frac{50 - 20}{50}\right) = \\ &= 1 - \Phi\left(\frac{3}{5}\right) = 1 - 0.7257 = 0.2743 \end{aligned}$$

In the air corridor

$$\begin{aligned} P(-50 \leq \xi \leq 50) &= P\left(\frac{-50 - 20}{50} \leq \frac{\xi - 20}{50} \leq \frac{50 - 20}{50}\right) = \\ &= \Phi\left(\frac{3}{5}\right) - \Phi\left(-\frac{7}{5}\right) = \Phi\left(\frac{3}{5}\right) - \left(1 - \Phi\left(\frac{7}{5}\right)\right) = 0.6449 \end{aligned}$$

Exercise 13.5 (Normal distribution MATLAB solution)

```
1-normcdf(3/5)
normcdf(3/5)-(1-normcdf(7/5))
%%% Simulation
n = 10^5;
x = normrnd(20,50,[1,n]);
length(x(x>50))/n
length(x(x>-50&x<50))/n
```

Exercise 15.4 (M2)

In a match making company the matchboxes are filled with an automatic machine. The number ξ of match-sticks in a randomly chosen box is a random variable with the following distribution:

number	47	48	49	50	51	52	53
probability	0.05	0.10	0.15	0.40	0.15	0.10	0.05

- a) Find the expected value of ξ .
- b) Find the variance of ξ .
- c) Find the value of $P(48 < \xi < 52)$.

a)

$$\mathbb{E}\xi = 47 \times 0.05 + 48 \times 0.1 + 49 \times 0.15 + 50 \times 0.4 + 51 \times 0.15 + 52 \times 0.1 + 53 \times 0.05 = 50$$

b)

$$\mathbb{D}^2\xi = \mathbb{E}(\xi - \mathbb{E}\xi)^2 = 9 \times 0.05 + 4 \times 0.1 + 1 \times 0.15 + 0 + 1 \times 0.15 + 4 \times 0.1 + 9 \times 0.05 = 2$$

c)

$$P(48 < \xi < 52) = P(\xi = 49) + P(\xi = 50) + P(\xi = 51) = 0.7$$

Literature

- ❑ Ágnes Baran: Mathematics for engineers 1. (Laboratory slides)
- ❑ Sándor Baran: Probability theory and statistics
- ❑ Fazekas István: Valószínűségszámítás, Debreceni Egyetemi Kiadó, 2009
- ❑ Fazekas István: Bevezetés a matematikai statisztikába, Debreceni Egyetemi Kiadó, 2009
- ❑ Matlab examples:
<https://www.mathworks.com/help/examples.html>